

Artificial Intelligence (CSA1711) — Assessment Tool 3: MCQ with Correct Answers

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment
 - b) An agent that uses a knowledge base and logical inference to decide actions ✓ Correct Answer**
 - c) An agent that only follows pre-defined scripts
 - d) An agent that learns from rewards
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Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$
 - b) $\neg P \vee Q$
 - c) $\neg Q \rightarrow \neg P$ only
 - d) Both b and c ✓ Correct Answer**
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Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions
 - b) A conjunction of disjunctions (clauses) ✓ Correct Answer**
 - c) A conjunction of literals only
 - d) A sequence of implications
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Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass
 - b) Every entity that is a student passes ✓ Correct Answer**
 - c) If something passes, it is a student
 - d) There exists at least one student who passes
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Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
 - b) Make two logical expressions syntactically identical ✓ Correct Answer**
 - c) Prove a goal using backward chaining
 - d) Derive facts using Modus Ponens
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Q6. In the Resolution algorithm, deriving the empty clause $\square$ indicates:

- a) The knowledge base is consistent
 - b) The negated goal is satisfiable
 - c) A contradiction — the original goal is proven ✓ Correct Answer**
 - d) More clauses need to be added
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Q7. Forward Chaining is best described as a _____ strategy.

- a) Goal-driven / top-down
 - b) Data-driven / bottom-up ✓ Correct Answer**
 - c) Heuristic-based depth-first
 - d) Probabilistic inference
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Q8. Backward Chaining begins reasoning from:

- a) All known facts toward a conclusion
 - b) The goal and works backward to verify supporting facts ✓ Correct Answer**
 - c) A random clause in the knowledge base
 - d) The resolution refutation tree
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Q9. Which step is NOT part of converting a sentence to CNF?

- a) Eliminate biconditionals ($\leftrightarrow$)
 - b) Move negations inward using De Morgan's laws
 - c) Apply Skolemization to remove existential quantifiers ✓ Correct Answer**
 - d) Distribute AND over OR
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Q10. Knowledge Engineering in FOL involves:

- a) Training a neural network on domain-specific data
 - b) Defining domain concepts, relations, and axioms formally in FOL ✓ Correct Answer**
 - c) Writing heuristic search strategies
 - d) Encoding reward functions for reinforcement learning
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Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

- a) $\neg Q$ is true
 - b) Q must be true ✓ Correct Answer**
 - c) P is uncertain
 - d) The KB is inconsistent
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Q12. The Most General Unifier (MGU) is the substitution that:

- a) Is the most specific possible match

- b) Makes two terms identical with minimal variable binding ✓ Correct Answer**
 - c) Eliminates all variables from a clause
 - d) Applies resolution to produce CNF
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Q13. Which inference method is more appropriate when a specific goal is known?

- a) Forward Chaining
 - b) Truth Table method
 - c) Backward Chaining ✓ Correct Answer**
 - d) A* search
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Q14. Propositional Logic differs from First Order Logic because it:

- a) Supports probabilistic reasoning
 - b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions ✓ Correct Answer**
 - c) Is always more expressive
 - d) Handles uncertainty better
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Q15. Resolution Refutation proves a goal G by:

- a) Directly deriving G from the KB using forward chaining
 - b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$) ✓ Correct Answer**
 - c) Checking G in the truth table
 - d) Applying Modus Tollens repeatedly
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Q16. Learning from observation in AI refers to:

- a) An agent receiving numerical rewards from the environment
 - b) Improving performance by examining examples or experience from the environment ✓ Correct Answer**
 - c) Acquiring knowledge by following explicit symbolic rules
 - d) Memorizing a fixed dataset without generalization
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Q17. Inductive learning is best described as:

- a) Deriving specific conclusions from general rules
 - b) Generalizing a hypothesis from specific training examples ✓ Correct Answer**
 - c) Memorizing training data exactly
 - d) Applying search algorithms to find optimal policies
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Q18. In a Decision Tree, what does each internal node represent?

- a) A class label / output decision

- b) A probability distribution over outcomes
 - c) A test on an attribute / feature ✓ Correct Answer**
 - d) A training data sample
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Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

- a) Requires large amounts of training data to generalize
 - b) Learns only from numeric data using regression
 - c) Uses prior domain knowledge to explain and generalize a single example ✓ Correct Answer**
 - d) Applies unsupervised clustering to find hidden patterns
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Q20. In statistical learning, a Naive Bayes classifier assumes:

- a) All features are conditionally dependent on each other given the class
 - b) The data must follow a uniform distribution
 - c) All features are conditionally independent given the class label ✓ Correct Answer**
 - d) Labels are determined by a reward function
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Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

- a) Using the EM algorithm to estimate missing values
 - b) Training a deep neural network with dropout regularization
 - c) Directly computing maximum likelihood estimates from observed data ✓ Correct Answer**
 - d) Applying reinforcement signals to update weights
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Q22. The Expectation-Maximization (EM) algorithm is used when:

- a) Training data is fully labeled and no variables are hidden
 - b) An agent explores an unknown environment using trial and error
 - c) The dataset contains hidden (latent) variables that cannot be directly observed ✓ Correct Answer**
 - d) Backpropagation fails to converge in a neural network
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Q23. In a Neural Network, the activation function serves to:

- a) Initialize the weights of each neuron to zero
 - b) Normalize the input dataset before training
 - c) Introduce non-linearity to enable learning of complex patterns ✓ Correct Answer**
 - d) Store previously learned examples in memory
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Q24. Which of the following correctly describes backpropagation in a neural network?

- a) Weights are updated in the forward pass using input gradients
- b) Outputs of each layer are stored and reused in later epochs

- c) **The error is propagated backward from output to input layers to adjust weights** ✓ Correct Answer
 - d) A separate model is trained for each hidden layer independently
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Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

- a) Minimizing the number of training examples used
 - b) Grouping similar data points into clusters using centroids
 - c) **Finding the maximum-margin hyperplane that separates classes** ✓ Correct Answer
 - d) Updating weights using Q-values from an environment
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Q26. The kernel trick in kernel machines (SVMs) allows:

- a) Reducing the number of support vectors needed
 - b) Accelerating gradient descent convergence
 - c) **Operating in a high-dimensional feature space without explicitly computing coordinates** ✓ Correct Answer
 - d) Replacing hidden layers in a deep neural network
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Q27. In Reinforcement Learning, the agent aims to maximize:

- a) Classification accuracy on a fixed test set
 - b) The number of features selected from the training data
 - c) **Cumulative reward received over time through interaction with the environment** ✓ Correct Answer
 - d) The size of the neural network hidden layer
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Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

- a) A supervised learning algorithm requiring labeled datasets
 - b) An unsupervised clustering technique using centroids
 - c) **A model-free, off-policy temporal difference learning method** ✓ Correct Answer
 - d) A decision-tree-based ensemble classifier
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Q29. Overfitting in machine learning occurs when:

- a) The model performs poorly on both training and test data
 - b) The learning rate is set too low during gradient descent
 - c) **The model fits training data too closely and generalizes poorly to new data** ✓ Correct Answer
 - d) The training set has more features than training examples
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Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

- a) RL uses labeled input-output pairs; SL uses reward signals
- b) Both RL and SL require explicit reward functions

c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples ✓ Correct Answer

d) RL is always faster to converge than SL on any dataset
